Tugas Pendahuluan Karakterisitik Sistem Orde Satu

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A.

Dik : $2y'(t) + 3y(t) = x(t)$ $\quad y(0) = 0$

Dit : • $\frac{Y(s)}{X(s)}$?

• gambar $y(t)$ jika $x(t) = 2$

Ubah ke transformasi Laplace

$2(sY(s) - y(0)) + 3Y(s) = X(s)$

$= 2(sY(s) - 0) + 3Y(s) = X(s)$

$= 2sY(s) + 3Y(s) = X(s)$

$= Y(s)(2s + 3) = X(s)$

Sehingga didapat $Y(s)/X(s)$

$$\frac{Y(s)}{X(s)} = \frac{1}{2s+3}$$

Bentuk umum sistem orde 1 :

$$\frac{Y(s)}{X(s)} = \frac{K}{\tau s + 1}$$

K : Penguatan

τ = konstanta waktu

ubah fungsi alih yang didapat jadi sesuai dengan bentuk umumnya :

$$\frac{Y(s)}{X(s)} = \frac{1}{2s+3} \times \frac{1/3}{1/3}$$

$$\frac{Y(s)}{X(s)} = \frac{1/3}{2/3 s + 1}$$

$$\frac{Y(s)}{X(s)} = \frac{0,34}{0,67s + 1}$$ $\quad K = 0,34$, $\tau = 0,67$

jika $x(t) = 2$, maka $y(\infty) = k \cdot 2$

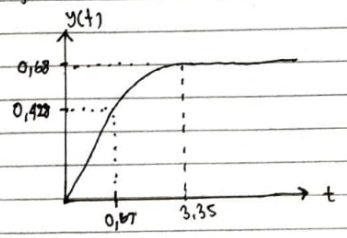
$= 0,34 \cdot 2 = 0,68$

$y(\tau) = y(\infty) \times 63\%$

$= 0,68 \times 63\% = 0,428$

$t_{steady\text{-}state} = 5\tau$

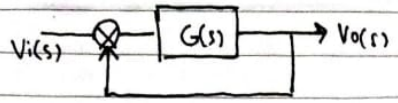
$= 5 \times 0,67 = 3,35$



SiDU

B.

Dik :



$G(s) = \frac{10}{2s+1}$ $\quad H(s) = 1$

Dit : Tentukan fungsi alih $Vo(s)/Vi(s)$

Gambar $Vo(t)$ jika $Vi(t) = 2$

Tentukan penguatan (k) dan konstanta waktunya (τ)

Menentukan fungsi alih :

$$\frac{Vo(s)}{Vi(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$\frac{Vo(s)}{Vi(s)} = \frac{\frac{10}{2s+1}}{1 + \frac{10}{2s+1}(1)} \times \frac{2s+1}{2s+1}$$

$$= \frac{10}{2s+1+10}$$

$$= \frac{10}{2s+11}$$

Ubah fungsi alih yang didapat agar sesuai dgn bentuk umumnya :

$$\frac{Vo(s)}{Vi(s)} = \frac{10}{2s+11} \times \frac{\frac{1}{11}}{\frac{1}{11}}$$

$$= \frac{\frac{10}{11}}{\frac{2}{11}s + 1}$$

$$= \frac{0,91}{0,18s+1}$$ $\quad K = 0,91$ $\quad \tau = 0,18$ detik

Jika $Vi(t) = 2$, maka :

$Vo(\infty) = k \cdot Vi(t)$

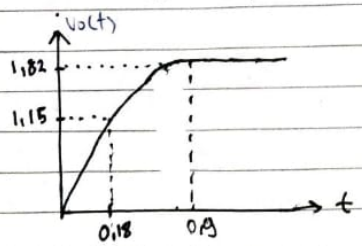
$= 0,91 \cdot 2 = 1,82$

$t_{steady-state} = 5\tau$

$= 5 \times 0,18 = 0,9$ s

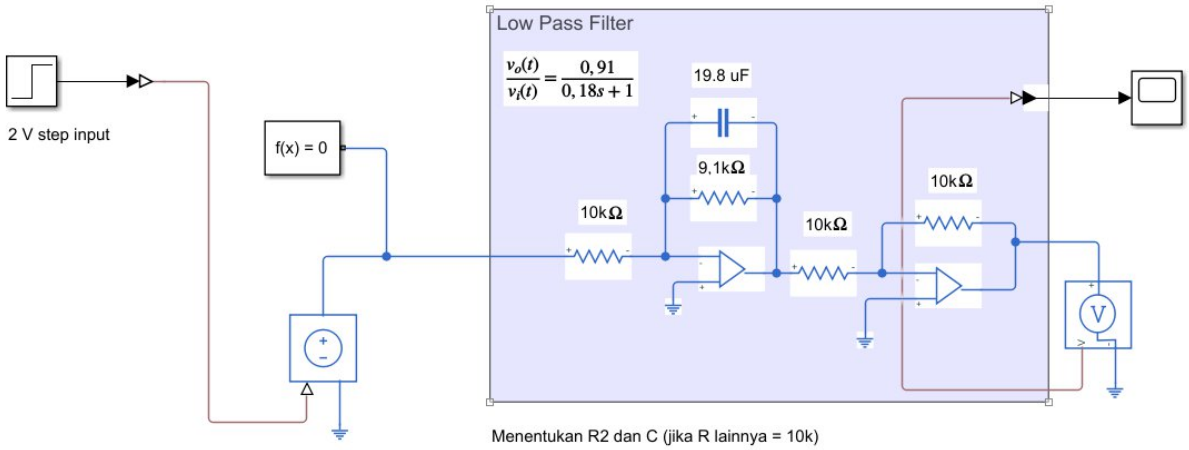
$Vo(\tau) = Vo(\infty) \times 63\%$

$= 1,82 \times 63\% = 1,15$



SIDU

Rangkaian simulasi:



Menentukan R2 dan C (jika R lainnya = 10k)

$$R_2 = K \times R_1$$

$$R_2 = 0,91 \times 10k\Omega = 9,1k\Omega$$

$$C = \frac{\tau}{R_2}$$

$$C = \frac{0,18}{9,1k} = 19,8\mu F$$

Output rangkaian:

